Who leaves teaching?

Teacher attrition in the United States, 2005–2025

EXECUTIVE SUMMARY

- Each year, **15.1%** of degree-holding school teachers leave the profession and do not return within our observation window. The conventional 12-month measure gives 17.6%, but one in six of those leavers is teaching again within three months.
- Persistent attrition rose from about 13% in 2005–2010 to about 17% in 2022–2024, with a peak in the pandemic year and a new high of 18% in 2024.
- Part-time work (+11 percentage points) and private-sector employment (+7) are the strongest predictors of leaving; higher pay retains. Risk is U-shaped in age, lowest around 40, and higher for Black and non-citizen teachers. A new baby during the year does not move men's exit probability but raises women's by 5 points, and most new mothers who leave exit the labor force entirely. The results are almost identical under both definitions.
- The typical leaver is a woman in her late career, working part-time, more often in a private school. The highest-risk decile has a 37% exit probability, 2.5 times the average.

15.1 %	**+11 pp**	**1 in 6**
of teachers leave the profession for good each year	extra exit risk from part-time work	12-month leavers is back teaching within 3 months

1. Data

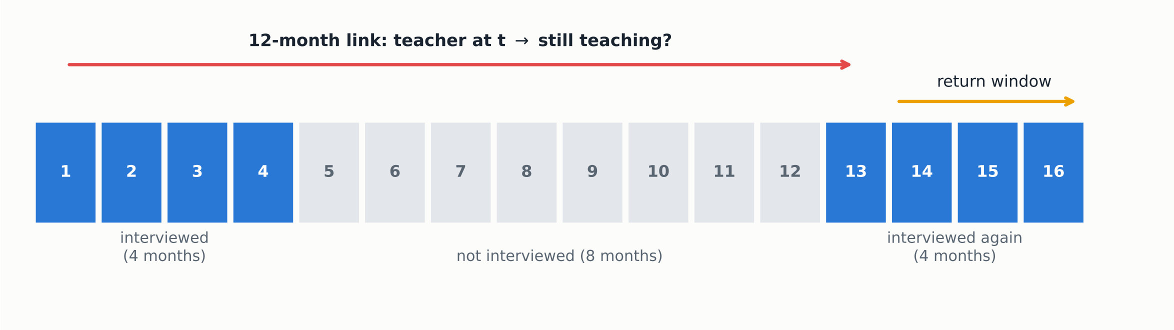
We use Current Population Survey (CPS) basic monthly microdata: 251 monthly files from January 2005 to December 2025, obtained from the U.S. Census Bureau (2024–2025) and the NBER mirror (2005–2023). October 2025 was never released.

The CPS interviews each household on a 4-8-4 rotation (Figure 1): four consecutive months in sample, eight months out, four months back in. A person is therefore observed for at most 16 months, and only once. Within that window, each adult interviewed in year t is re-interviewed in the same calendar month of $t+1$. We link the two interviews with household and person identifiers and validate each match on sex, race and age progression. Nobody can be followed across several years, so each base year is a fresh cohort and every rate in this brief is computed wave by wave.

Table 1 shows the population funnel, pooling all 251 months.

Table 2 describes the variables available for each linked teacher, separately for stayers and per-

Figure 1: The CPS follows each person for at most 16 months. Interview calendar of one rotation group; numbers are months since first interview.



Notes: A household entering the survey is interviewed in four consecutive months, rests for eight, and is interviewed again in four more. The 12-month link compares an interview from the first block with the interview held exactly one year later; the return window uses the remaining interviews of the second block.

Table 1: From the CPS universe to the analytic sample.

	Persons	Population represented
Adults interviewed in an average month	94,630	240.6M
employed	57,572	148.1M
school teachers (occ. 2300–2330)	2,094	5.2M
with bachelor's degree or higher	1,849	4.6M
Teachers linked to their interview 12 months later (unique persons, 20 waves)	174,872	the same 4.6M
with re-interviews after $t+12$: **main sample**	129,897	the same 4.6M

Notes: Teachers are employed persons whose main job is teaching at the preschool or kindergarten, elementary or middle school, secondary school, or special education level (Census occupation codes 2300 to 2330, which are identical across the 2002, 2010 and 2020 classifications) and who hold at least a bachelor's degree. Only persons in the first four months of their rotation can be linked one year ahead. Losses occur because the survey samples addresses and does not follow people who move, but they are not selective against teachers: 78 percent of teachers are found and validated twelve months later, against 74 percent of other college-educated workers and 71 percent of all employed, a ranking that holds in every single year. Linked teachers keep their population weights, so weighted rates remain representative of the 4.6 million teachers. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

sistent leavers. The raw differences already point at the profile of the leaver: older, more often part-time, less credentialed, with fewer young children at home.

Defining a leaver. The conventional measure, used by the NCES Teacher Follow-up Survey, counts anyone not employed as a school teacher twelve months after baseline. Temporary exits such as parental leave or a short unemployment spell inflate it: 15.7% of those leavers are teaching again within three months. Our main measure is therefore the persistent leaver: out of teaching at $t+12$ and in every later re-interview. It is computed on the 129,897 teachers whose rotation allows at least one re-interview. Exits longer than a year that end in a return cannot be detected, so persistent attrition is still an upper bound on permanent exit, just a much tighter one. All rates use CPS person weights.

Caveat. Occupation in the returning rotation is re-coded independently, which inflates measured switching. Levels should be read as upper bounds; comparisons across groups and over time, the object of this brief, are far less affected.

Table 2: Descriptive statistics of the analytic sample (weighted means).

	All teachers	Stayers	Persistent leavers	Difference
Age, years	44.0	43.5	46.7	+3.15***
Female	76.8%	76.8%	77.0%	+0.2 pp
Female aged 25–44	37.8%	38.8%	31.7%	-7.1 pp***
Married	72.0%	72.5%	69.5%	-3.0 pp***
Number of own children (<18) at home	0.8	0.9	0.6	-0.22***
Child under 6 at home	18.1%	18.9%	13.9%	-5.0 pp***
New baby during the year	2.5%	2.5%	2.8%	+0.3 pp
Black	8.1%	7.5%	11.6%	+4.2 pp***
Hispanic	7.9%	7.8%	8.5%	+0.7 pp*
Non-citizen	1.7%	1.5%	3.0%	+1.5 pp***
Master's degree or higher	52.9%	53.7%	48.4%	-5.3 pp***
Professional degree or doctorate	2.8%	2.5%	4.4%	+2.0 pp***
Part-time (<35 h/week)	9.0%	7.2%	19.1%	+11.9 pp***
Holds more than one job	8.1%	7.9%	9.4%	+1.5 pp***
Public-sector employer	78.3%	80.5%	65.7%	-14.9 pp***
Family income $75k+	76.4%	77.3%	71.4%	-5.9 pp***
Elementary / middle school	61.8%	61.9%	61.2%	-0.7 pp
Preschool / kindergarten	7.7%	7.1%	10.6%	+3.5 pp***
Secondary school	23.1%	23.5%	20.5%	-3.0 pp***
Special education	7.5%	7.5%	7.6%	+0.1 pp
Persons	129,897	110,720	19,177	

Notes: The sample is all linked teachers whose rotation includes at least one interview after the twelve-month link. All variables are measured at baseline and weighted with CPS person weights. Children are own children under eighteen living in the household at the time of the interview. The last column reports the stayer-leaver difference and the significance of a two-sample test with standard errors clustered by household: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

2. The teacher workforce

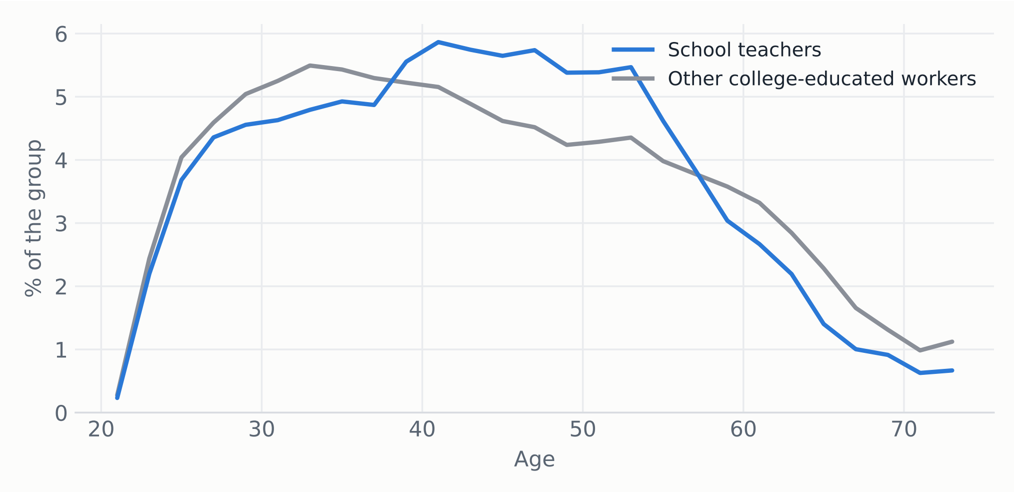
Before asking who leaves, this section describes who teaches, comparing teachers with all other college-educated workers in 2021–2025. Teaching is a demographically distinct segment of the graduate labor market. Teachers are overwhelmingly women (76 vs 49 percent) and disproportionately white; Asian and foreign-born graduates are strongly under-represented. They are slightly younger on average, more often married and with more children at home. Their age distribution (Figure 2) is more concentrated in mid-career, with a thicker mass between 35 and 55 and thinner tails at both ends.

As a job, teaching is even more distinctive (Figure 3 and Table 3). Three quarters of teachers work for the public sector, against 15 percent of comparable graduates, and 47 percent are union members, against 8 percent. Half hold a master's degree, twice the rate of other graduates. Hours are standard full-time and part-time work is less common than elsewhere. The price of this stability is pay: the median teacher earns $1,200 a week, 20 percent less than the $1,500 of comparable college-educated workers. Geographically, teachers are spread across the country like the rest of the college workforce, so attrition patterns are not driven by where teachers live.

Weighted to population, the CPS counts a stable stock of about 4.7 million degree-holding school teachers (Figure 4), most of them in elementary and middle school (Figure 5).

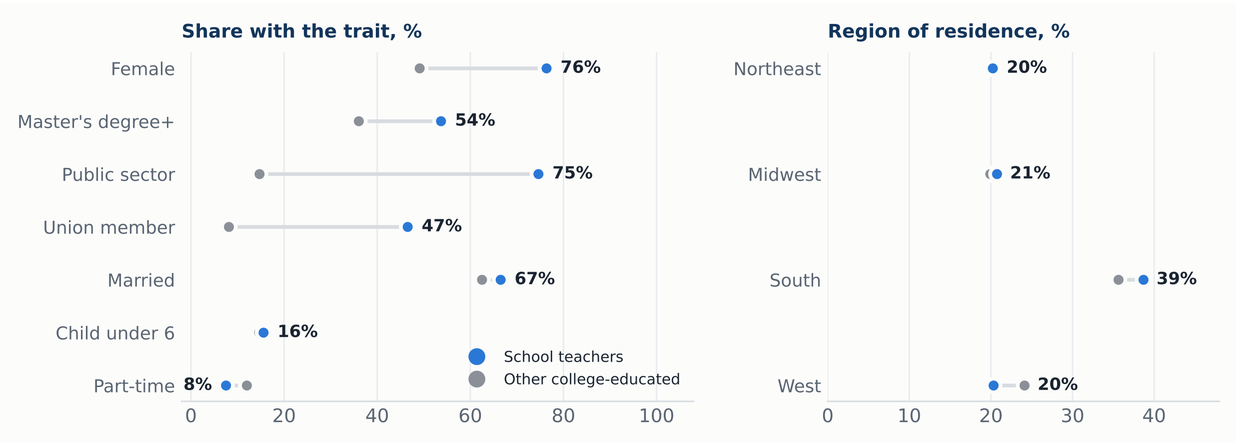
The linked pairs also show both sides of turnover (Figure 6). Teaching loses and recruits 15 to 20% of its stock every year. It is a high-churn occupation, and in recent years exits have run above entries. Most persistent leavers move to another occupation (9.9 of the 15.1 points); 4.5

Figure 2: Age distribution of school teachers and of other college-educated workers, 2021–2025.



Notes: Both groups are employed persons with at least a bachelor's degree; teachers are identified as in Table 1. Weighted to population. Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

Figure 3: Composition of the teaching workforce vs other college-educated workers, 2021–2025.



Notes: Same groups as Figure 2. Union membership is only asked in the outgoing rotations. Shares are weighted to population. Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

points leave the labor force and 0.6 become unemployed.

3. Attrition over time

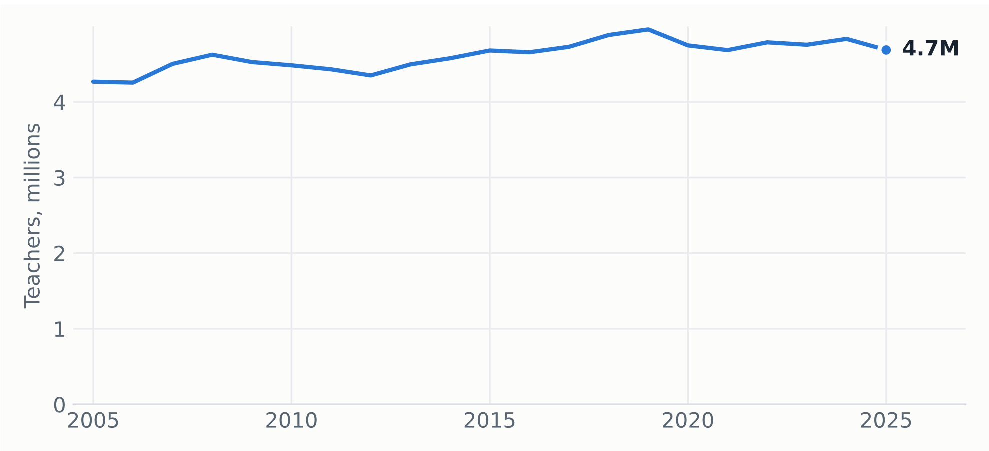
One in six 12-month leavers is teaching again within three months (women 16.1%, men 14.0%). These are temporary exits that the persistent definition strips out. Net of this churn, persistent attrition still climbed from about 13% in 2005–2010 to about 17% in 2022–2024 (Figure 7), peaking in the pandemic year (18.4% in 2019→2020) and reaching a new high of 18.0% in 2024. Gender gaps are small and unstable; in the most recent year men (19.8%) exceeded women (17.4%).

Table 3: Balance: school teachers vs other college-educated workers, 2021–2025 (weighted).

	School teachers	Other college-educated workers	Difference
Age, years	43.5	43.9	-0.44***
Female	76.4%	49.1%	+27.2 pp***
White	83.8%	75.4%	+8.5 pp***
Black	9.8%	10.2%	-0.4 pp
Asian	3.9%	11.7%	-7.7 pp***
Hispanic	10.5%	10.6%	-0.1 pp
Non-citizen	2.5%	7.7%	-5.2 pp***
Married	66.5%	62.5%	+4.0 pp***
Number of own children (<18) at home	0.8	0.6	+0.19***
Child under 6 at home	15.6%	14.7%	+0.9 pp***
Master's degree	50.4%	25.8%	+24.6 pp***
Professional degree or doctorate	3.3%	10.3%	-7.0 pp***
Public-sector employer	74.6%	14.7%	+59.9 pp***
Union member[a]	46.6%	8.2%	+38.4 pp***
Usual weekly hours	40.2	39.8	+0.40***
Part-time (<35 h/week)	7.5%	12.0%	-4.5 pp***
Holds more than one job	7.3%	6.2%	+1.2 pp***
Weekly earnings, median[a]	$1,200	$1,500	−$300***
Family income $75k+	85.3%	85.5%	-0.2 pp
Persons (monthly interviews pooled)	85,952	1,059,720	

Notes: Both groups are employed persons with at least a bachelor's degree. The last column reports the teacher minus non-teacher difference and the significance of a two-sample test with standard errors clustered by household: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. [a] Union membership and weekly earnings are only asked in the outgoing rotations (months in sample 4 and 8). Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

Figure 4: The stock of school teachers. Employed school teachers with a bachelor's degree or higher, annual average of monthly estimates.

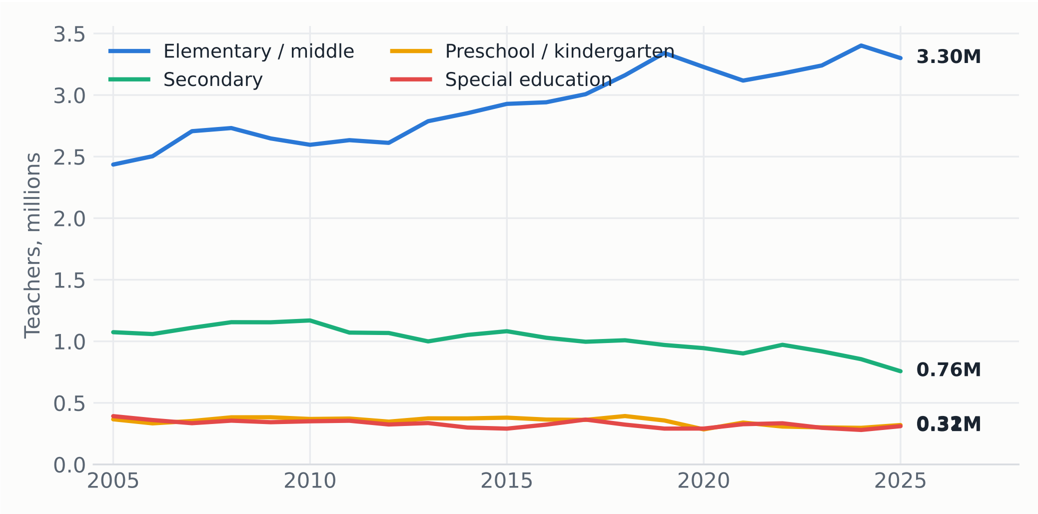


Notes: Teachers are employed persons whose main job is school teaching (Census occupation codes 2300 to 2330) and who hold at least a bachelor's degree, weighted to population. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

4. Where leavers go

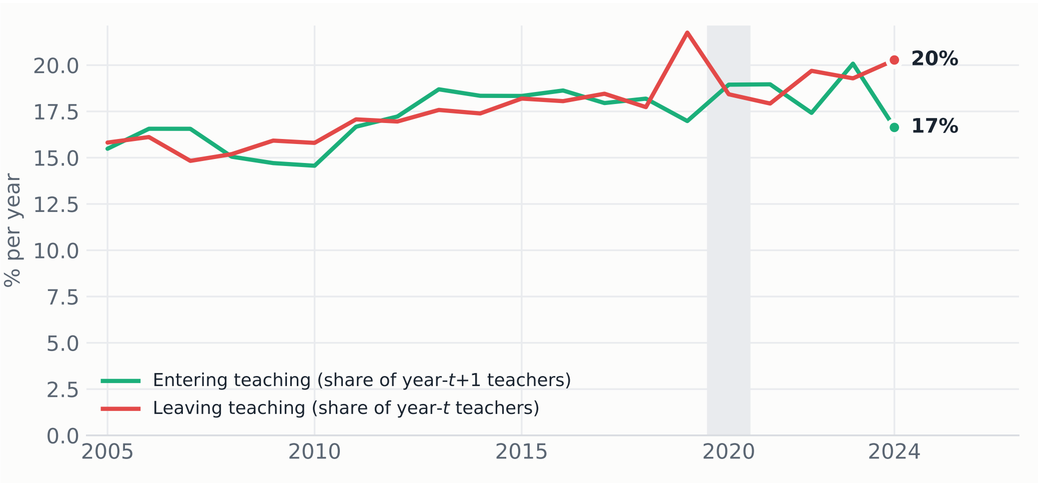
The linked design tells apart those who abandon the labor force from those who change occupation, and for the latter it records the destination occupation (Figure 8). Three facts stand out. First, the single largest destination, 30 percent, is another education job: postsecondary teaching, tutoring and instruction, teaching assistance, librarianship or school administration. A large

Figure 5: Teachers by teaching level.



Notes: Same identification as in Figure 4, split by teaching level. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

Figure 6: Teachers entering and leaving each year. Gross annual flows in the linked pairs, 12-month definitions.



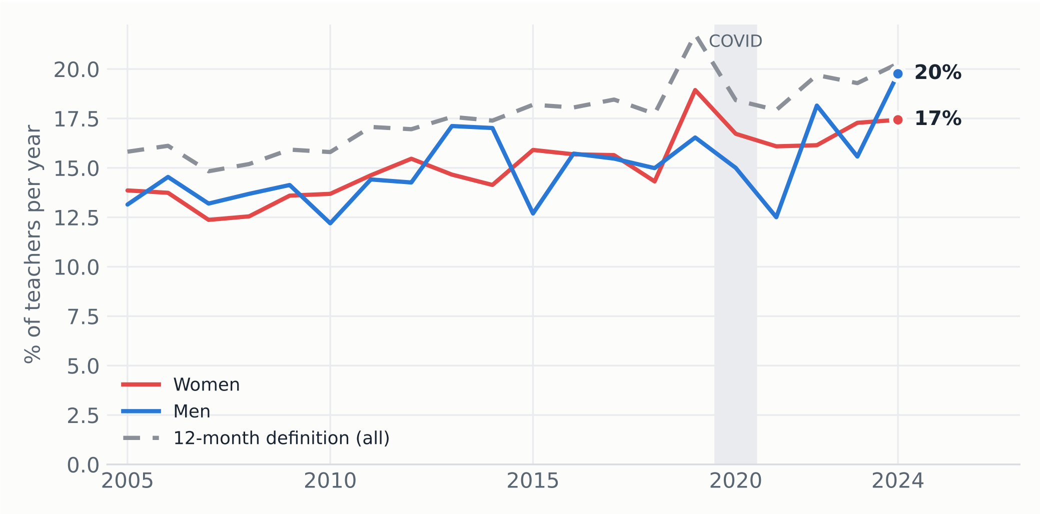
Notes: A person enters teaching if she is a degree-holding school teacher at the second interview but was not one a year earlier; leaving is the reverse. Because the linked sample misses people who move, both flows are conservative. Source: authors' calculations from linked CPS pairs, 2005–2025.

share of exits from classroom teaching are therefore not exits from education. Second, retirement accounts for 17 percent (mean age 62), and only 4 percent are unemployed a year later. Third, the gender contrast is sharp exactly where the family results of the model point: 14 percent of women who leave are out of the labor force for reasons other than retirement or disability (mean age 39, consistent with family care), against 5 percent of men, while men move more often into management and professional jobs (26 vs 19 percent).

5. Who leaves

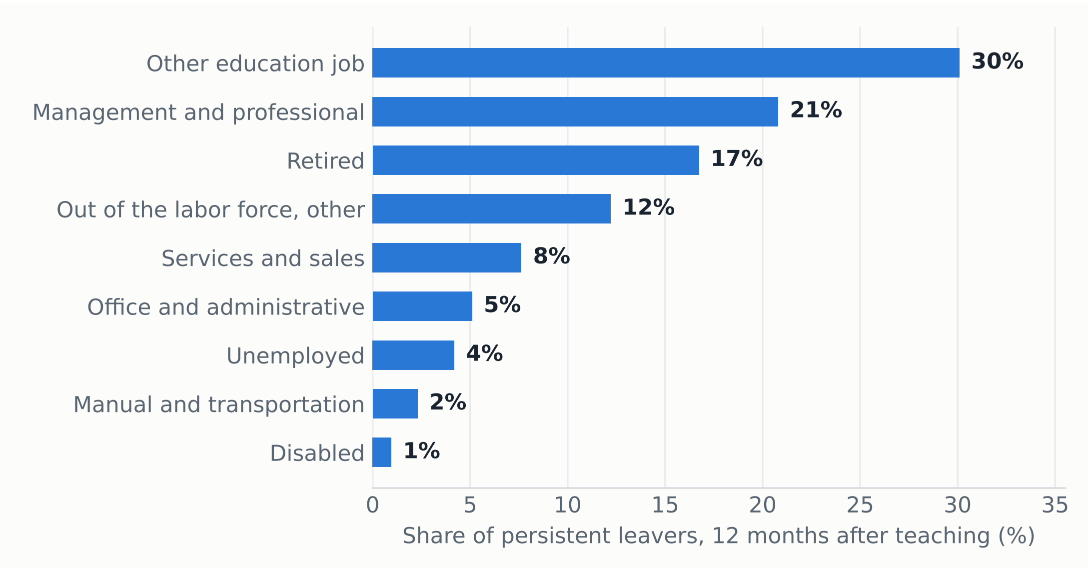
Attrition concentrates in three margins: hours, teaching level and, more weakly among degree holders, credentials. Part-time teachers leave at more than twice the rate of full-time teachers, and preschool/kindergarten is the most exposed level (Figure 10).

Figure 7: Persistent attrition by year and gender.



Notes: The return window is capped by the survey design at three monthly interviews after the twelve-month link; 21,873 of the 30,244 leavers are re-interviewed at least once. Source: authors' calculations from linked CPS pairs, 2005–2025.

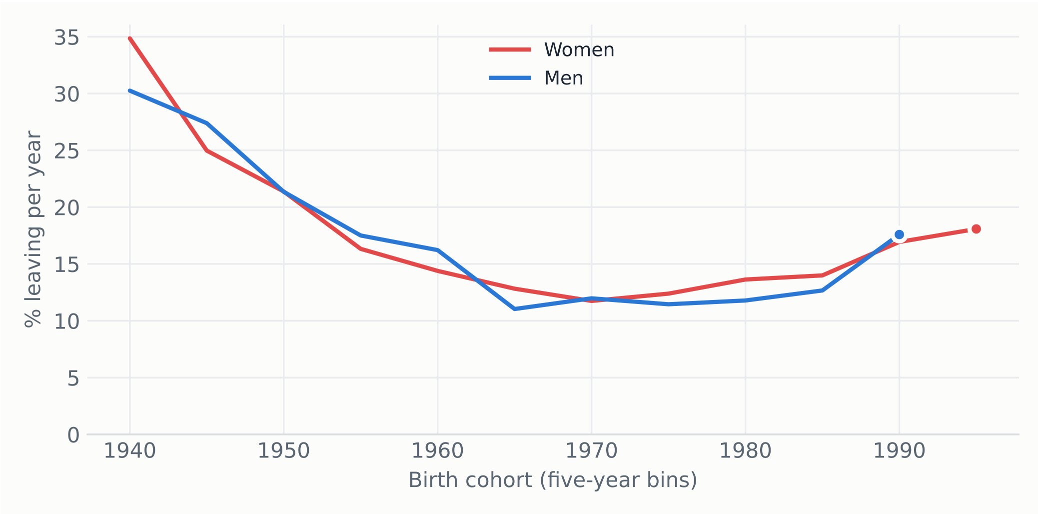
Figure 8: Situation of persistent leavers twelve months after teaching.



Notes: Education jobs other than school teaching include postsecondary teaching, tutoring and instruction, teaching assistance, librarians and education administration. The remaining employed leavers are grouped into broad occupation classes. Retirement, disability, unemployment and other non-participation come from the labor force status recorded at the second interview. Source: authors' calculations from linked CPS pairs, 2005–2025.

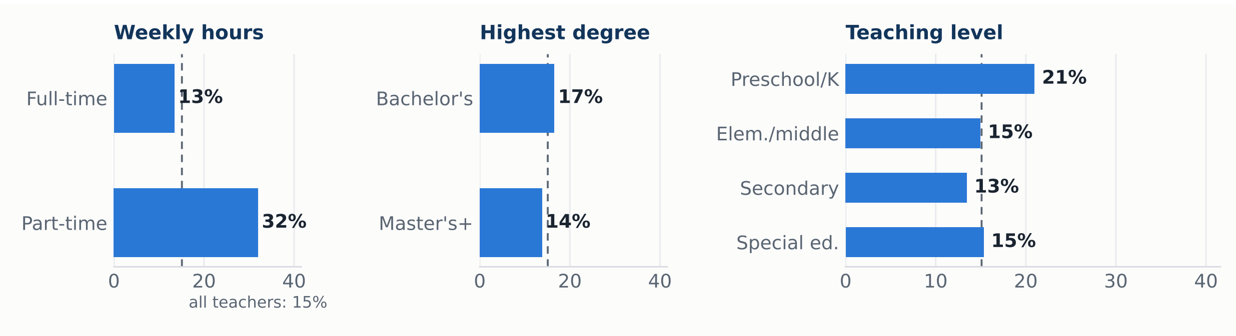
The lifecycle shape is not an artifact of how covariates enter the model. Plotting raw attrition by birth cohort separately for men and women (Figure 9) shows the same U: cohorts born before 1950, observed near retirement, leave at 25 to 35 percent per year; cohorts born in the 1960s and 1970s, observed mid-career, at 11 to 13 percent; and the youngest cohorts, observed at career entry, climb back to 18 percent. Men and women trace nearly the same curve, which anticipates the small conditional gender effect in the model.

Figure 9: Attrition by birth cohort and sex.



Notes: Weighted persistent-attrition rate by five-year birth cohort, pooling all twenty waves; each cohort is observed at the ages it reaches during 2005–2025, so the curve mixes age and cohort variation. Cells with fewer than 400 observations are omitted. Source: authors' calculations from linked CPS pairs, 2005–2025.

Figure 10: Persistent attrition by baseline characteristics, 2005–2025.



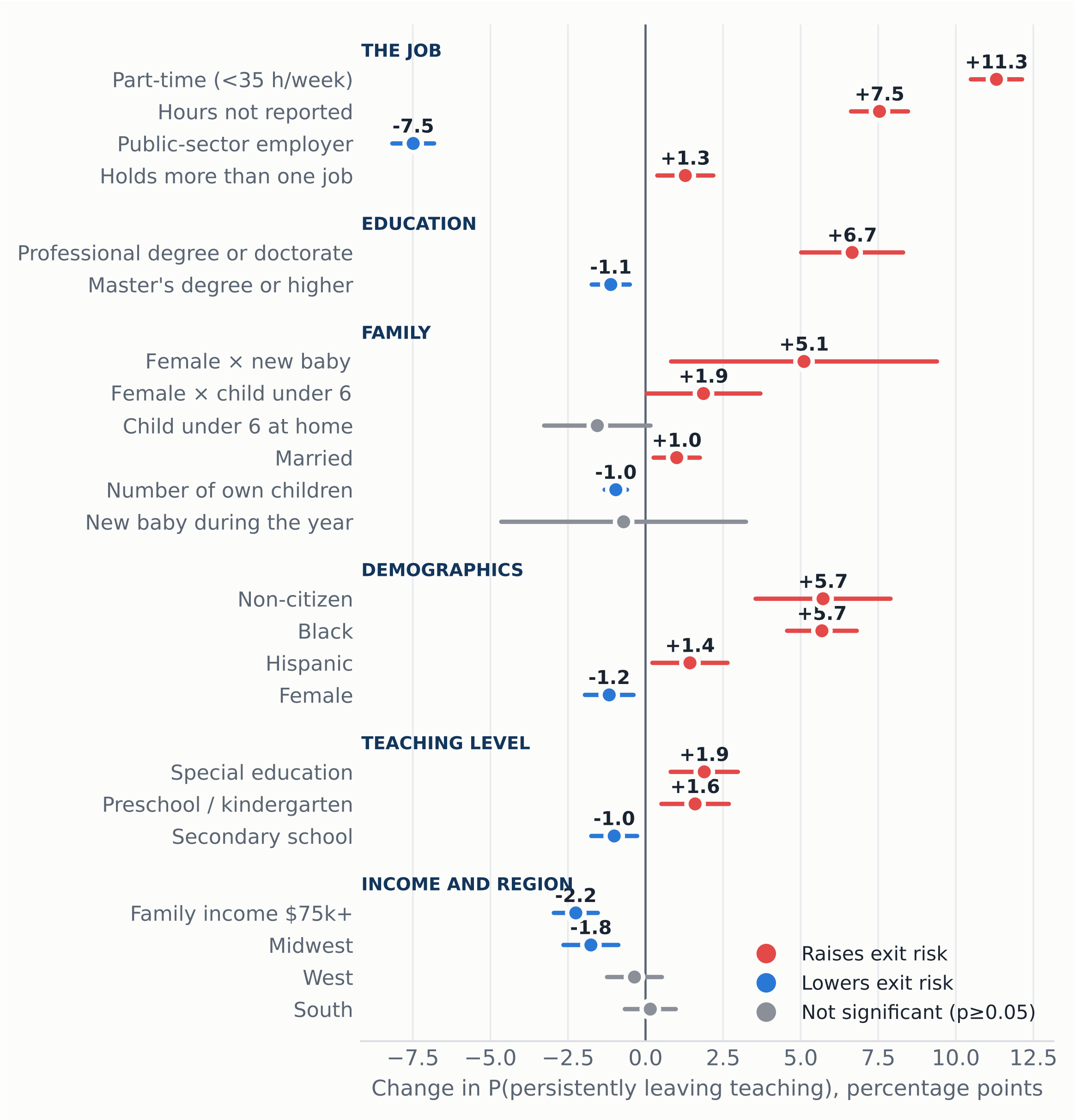
Notes: The dashed line marks the all-teacher average of 15.1 percent. Source: authors' calculations from linked CPS pairs, 2005–2025.

6. The model

We estimate a probit for the probability of persistently leaving, with covariates measured at baseline, base-year fixed effects and standard errors clustered by household (Table 4). Figure 11 plots the average marginal effects.

Five results stand out. First, the job margin dominates: part-time status raises exit probability by 11.3 points and private-sector employment by 7.5 points relative to public schools, the two largest effects in the model. Second, pay retains. Weekly earnings are only collected in the outgoing rotations, so we estimate the wage effect on that subsample (44,823 teachers): ten percent higher weekly earnings lower the probability of leaving by 0.3 points. Third, the family results separate stocks from events. The stock of children retains: each own child at home lowers exit risk by about 1 point. The event of a birth does the opposite, and only for women: a new baby during the year leaves men's exit probability unchanged (-0.7 points, not significant) but raises women's by 5.1 points, and 58 percent of the new mothers who leave exit the labor force entirely, against 30 percent of the average leaver. This is the fertility channel the literature emphasizes, invisible when only the stock of children is used. Fourth, minority and non-citizen teachers are significantly more likely to leave (Black $+5.7$, non-citizen $+5.7$, Hispanic $+1.4$ points), a retention gap that matters for workforce diversity. Fifth, credentials cut both ways: relative to a bachelor's degree, a master's retains (-1.1) but a professional degree or doctorate strongly predicts leaving $(+6.7$ points), consistent with better options outside the classroom, including postsecondary teaching. Every effect is nearly identical under the conventional 12-month definition (part-time: $+13.6$ vs $+11.3$), so the predictors do not depend on how leaving is measured.

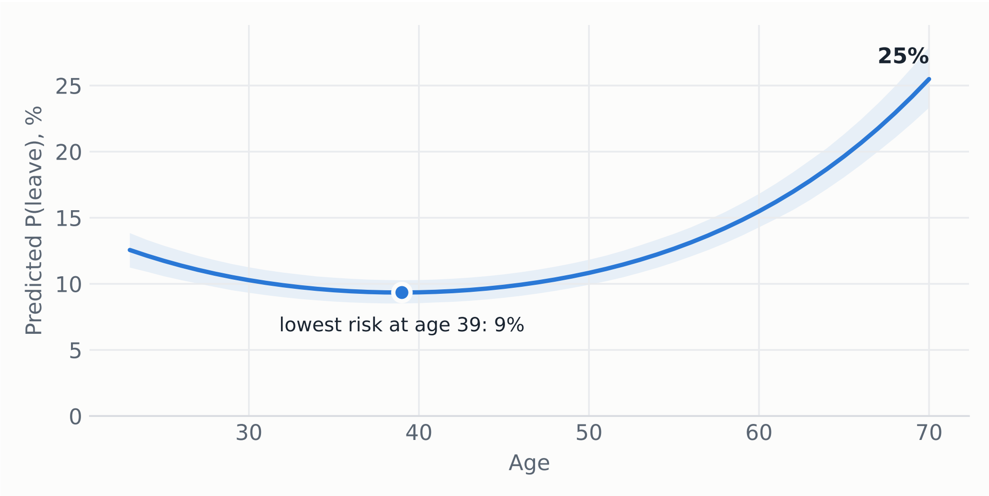
Figure 11: Average marginal effects on the probability of persistently leaving teaching, with 95% confidence intervals.



Notes: Estimates from the probit of Table 4. The age profile is shown separately in Figure 12.

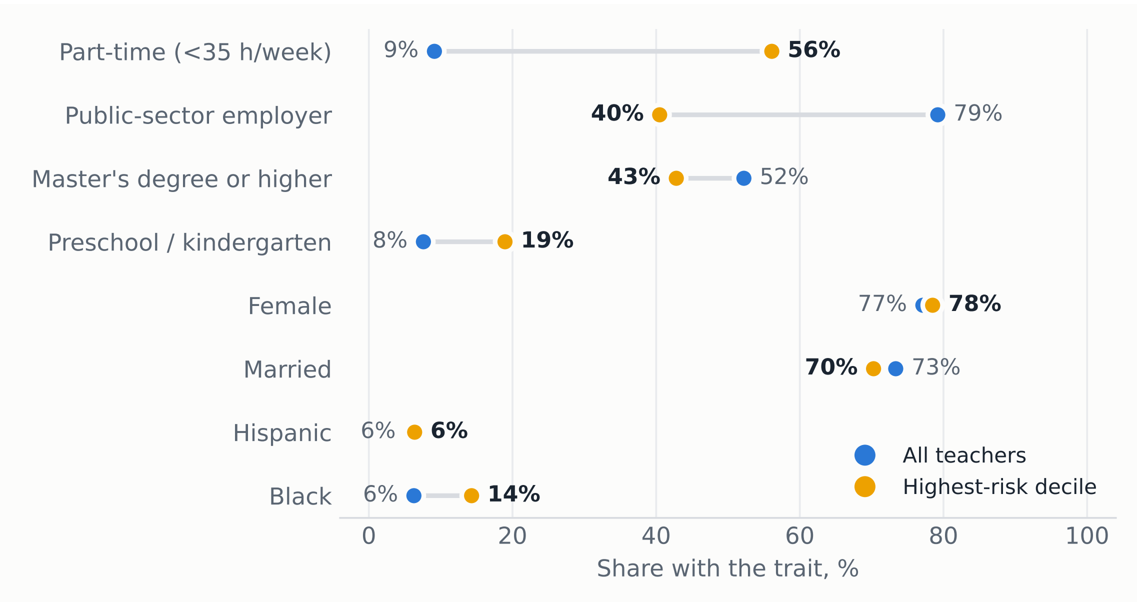
Age enters with a robust U-shape (Figure 12). Predicted risk falls to a minimum of 9% around age 39, accelerates after 55 as retirement approaches, and reaches 25% at 70.

Figure 12: Predicted probability of leaving by age, other covariates at sample means. Shaded band: 95% confidence interval.



Notes: Predicted probabilities from the probit of Table 4, holding all other covariates at their sample means. The confidence band is simulation based.

Figure 13: Characteristics of the top predicted-risk decile vs all teachers.



Source: predicted probabilities from the probit of Table 4.

7. The typical leaver

Ranking teachers by predicted risk, the top decile has a mean exit probability of 37%, against 15% for the average teacher. It is overwhelmingly female (79%), late-career (mean age 55), six times more likely to work part-time (56% vs 9%), and only 40% work in a public school against 78% of all teachers, with Black teachers over-represented (14% vs 6%).

Table 4: Probit estimates: probability of persistently leaving teaching, 2005–2025.

	Probit coef.	(SE)	AME (pp)
Age (years)	-0.053***	(0.004)	-1.14***
Age^2/100	0.068***	(0.004)	+1.46***
Female	-0.055***	(0.019)	-1.17***
Married	0.047***	(0.018)	+1.01***
Number of own children	-0.045***	(0.009)	-0.96***
Child under 6 at home	-0.072*	(0.041)	-1.55*
Female × child under 6	0.087**	(0.044)	+1.87**
New baby during the year	-0.033	(0.094)	-0.71
Female × new baby	0.238**	(0.102)	+5.11**
Black	0.265***	(0.027)	+5.68***
Hispanic	0.067**	(0.029)	+1.43**
Non-citizen	0.267***	(0.052)	+5.72***
Master's degree or higher	-0.052***	(0.015)	-1.12***
Professional degree or doctorate	0.311***	(0.039)	+6.66***
Part-time (<35 h/week)	0.528***	(0.020)	+11.31***
Hours not reported	0.352***	(0.022)	+7.54***
Holds more than one job	0.060***	(0.021)	+1.28***
Public-sector employer	-0.349***	(0.016)	-7.49***
Family income $75k+	-0.105***	(0.017)	-2.25***
Midwest	-0.082***	(0.021)	-1.76***
South	0.007	(0.020)	+0.15
West	-0.017	(0.021)	-0.36
Preschool / kindergarten	0.075***	(0.026)	+1.60***
Secondary school	-0.047***	(0.018)	-1.01***
Special education	0.088***	(0.026)	+1.89***
Constant	0.094	(0.096)	
Base-year fixed effects		Yes	
Observations		129,897	
Pseudo R^2		0.068	

Notes: The outcome is the persistent leaver defined in Section 1. Standard errors, in parentheses, are clustered by household. The AME column reports average marginal effects in percentage points. The sample is all school teachers with a bachelor's degree or higher whose rotation includes at least one interview after the twelve-month link. Reference categories are full-time hours, a bachelor's degree, a private-sector employer, elementary or middle school, and the Northeast region. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

8. Policy implications

The results locate attrition in weak attachment to the job (part-time hours, private-sector positions, both career extremes) rather than in broad demographics, and show a rising trend since the early 2010s. Four levers follow. First, converting involuntary part-time positions to full-time attacks the single largest risk factor, and the wage result implies that pay raises retain, though modestly. Second, the private sector is where teaching careers break: private-school teachers leave at far higher rates, and the preschool exposure partly reflects that composition. Third, the Black, Hispanic and non-citizen retention gaps deserve monitoring, since they compound into the diversity of the future workforce. Fourth, with one in six leavers back within three months, cheap re-entry pathways such as fast-track rehiring, credential portability and return bonuses can recover part of the outflow at low cost.

Methods in one line. Twenty linked CPS 4-8-4 rotation panels, 2005→2025 · 174,872 teachers (129,897 with follow-up) · persistent-leaver outcome · probit, year FE, household-clustered SE · fully reproducible: `src/02_build_cps_panel.py` → `src/05_evolution.py` → `src/03_model_attrition.py`.